

Gaussian Processes for Dynamic Portfolio Optimization

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Context within the Summer School

Dynamic optimization in economics and finance

- Different **economic settings**:
 - Partial vs. general equilibrium
 - Continuous vs. discrete time
 - Complete vs. incomplete markets
- Different **optimality conditions**
 - Hamilton-Jacobi-Bellman (HJB) and Bellman equations
 - Euler equations/first-order conditions
 - Static optimality conditions in complete markets
- In general equilibrium, **market clearing** couples optimality conditions, additionally inducing **fixed-point structures**

⇒ Different settings naturally support **different ML solution methods**

Deep Equilibrium Nets

- Neural networks (NNs) approximate **equilibrium policy functions**
- Training minimizes **equilibrium residuals** over simulated states
- Accurate learning requires **extensive exploration** of the exogenous state space
- Requires **cheap** and **scalable simulation** for:
 - **generating** economically relevant states
 - **repeatedly evaluating** equilibrium (optimality + market clearing) conditions
- **Indirect-Inference-like learning:**
 - model simulation generates the training data, while equilibrium conditions provide the **auxiliary optimization criterion** for learning the NN parameters

Physics-Informed Neural Networks

- Naturally suited to **continuous-time economic models**
- Optimality/equilibrium characterized by **ODEs/PDEs** (e.g. HJB, Kolmogorov Forward, Master equations)
- NNs approximate **unknown solution functions**
- Training minimizes **interior equation residuals** and **boundary conditions** at collocation points:
 - Derivatives via **automatic differentiation**
 - **Mesh-free** (direct state-space sampling) and **scalable**
- In heterogeneous-agent models, NNs can additionally provide **low-dimensional representations** of **distributional states** entering Master equations

Dynamic optimization with Gaussian Processes

- Deep Equilibrium Nets/PINNs work well when optimality conditions admit **tractable representations** and can be **cheaply evaluated** at many states
- Many models instead feature **costly solution evaluations** (expectations, optimization, simulation) and/or no convenient **differential-equation characterization**
- Key need for **sample-efficient learning** from few expensive evaluations:
 - **Gaussian Processes** combine flexible function approximation with tractable **uncertainty quantification**
 - **Active learning** enables **adaptive sampling**, targeting regions of greatest solution uncertainty.

Discrete-time dynamic portfolio choice

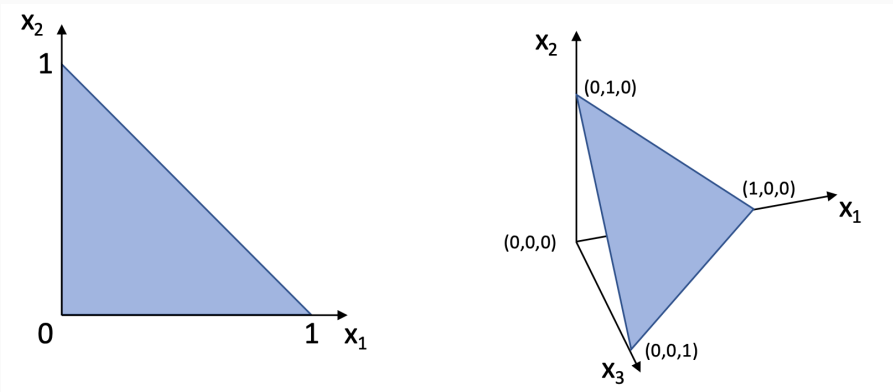
- Discrete-time intertemporal utility maximization under dynamic budget constraints
- Naturally characterized by a Bellman equation/recursion:
 - ⇒ Approximation errors can propagate across investment horizons
- Closed-form solutions require stringent assumptions on utility, return dynamics, market completeness and trading costs
- Otherwise, each recursion requires repeated expectation computation + optimization over the state space:
 - ⇒ Natural setting for sample-efficient function approximation

Dynamic Portfolio Choice with Transaction Costs

Transaction costs raise the challenge

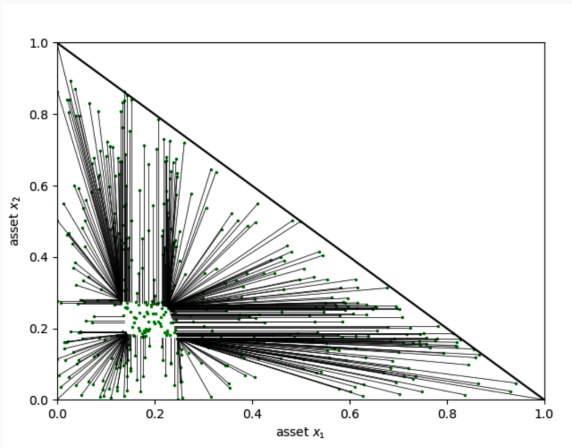
- Budget constraint with state dimension **not smaller** than number of risky assets:
 - **Curse of dimensionality** becomes more severe
 - **No closed-form solution** even in benchmark IID models (Constantinides, 1979, 1986)
 - Existing numerical methods hardly applicable beyond **3–4 risky assets**
- **Distinct trading regimes** over **endogenous irregularly shaped** state space:
 - Varying **smoothness** and **nonlinearity**, with non-differentiabilities at regime boundaries
 - Challenging for **global function approximators**, since approximation effort must **adapt** across regimes and near their boundaries

Feature 1: Curse of dimensionality and irregularly-shaped domain



- Budget constraint depends on D -dimensional vector of risky assets' wealth shares
- Irregularly-shaped domain with, e.g., no-short selling/borrowing constraints
- A challenge for grid-based function approximation methods

Feature 2: Intrinsically "distinct" portfolio behavior inside admissible domain



- "Distinct" portfolio behavior inside/outside endogenous no-trade region (NTR)
- Naturally varying smoothness properties in neighborhoods of boundary of NTR

Objective

A [comprehensive machine learning solution framework](#) for dynamic portfolio choice with proportional transaction costs:

- [Scalable](#)
- Accomodating [nonlinearities](#), [nondifferentiabilities](#) and [irregularly-shaped](#) domains
- "[Tunable-smooth](#)" over different subregions of the domain
- Allowing a [quantification](#) of the uncertainty of computed solutions
- [Extendable](#), e.g., to general equilibrium and state-dependent opportunity sets/transaction costs

Benchmark Setting and Key Issues

Dynamic optimal consumption/investment problem

- Trading dates $t \in \{0, 1, \dots, T\}$ and IID asset returns (R_f, R_{t+1})
- State variables $(W_t, \mathbf{x}_t) \in \mathcal{D}_t$ for wealth and risky assets' wealth fractions
- Consumption/investment feedback controls $(C_t, \boldsymbol{\delta}_t) \in \mathcal{D}(W_t, \mathbf{x}_t)$
- Time-additive optimal consumption investment problem:

$$V(W_0, \mathbf{x}_0) := \max_{(C, \boldsymbol{\delta})} \mathbb{E} \left[\sum_{t=1}^T u(t, C_t) \middle| W_0, \mathbf{x}_0 \right] \quad (1)$$

- Not everywhere differentiable dynamic budget constraint:

$$(W_{t+1}, \mathbf{x}_{t+1}) = BC(W_t, \mathbf{x}_t, C_t, \boldsymbol{\delta}_t; R_f, R_{t+1}) \quad (2)$$

Case I: Not differentiable dynamic budget constraint with proportional TC

- Risky assets' **wealth fraction** dynamics:

$$\mathbf{x}_{t+1} = \frac{W_t}{W_{t+1}} (\mathbf{x}_t + \boldsymbol{\delta}_t) \odot \mathbf{R}_{t+1}$$

- **Wealth** dynamics with homogeneous **proportional transaction cost (PTC)** $\tau \geq 0$:

$$W_{t+1} = (W_t - C_t - \mathbf{1}^\top (\mathbf{x}_t + \boldsymbol{\delta}_t + \tau |\boldsymbol{\delta}_t|) W_t) R_f + W_t (\mathbf{x}_t + \boldsymbol{\delta}_t)^\top \mathbf{R}_{t+1}$$

- $W_t - C_t - \mathbf{1}^\top (\mathbf{x}_t + \boldsymbol{\delta}_t + \tau |\boldsymbol{\delta}_t|) W_t$ equals **saving net of** risky investment and TC
- PTC induces **not differentiable** wealth dynamics
- For $\tau = 0$, wealth dynamics only depends on **"aggregate control"** $\mathbf{w}_t := \mathbf{x}_t + \boldsymbol{\delta}_t$, i.e., the relevant state space does **not scale** with the number of risky assets

Case II: Irregularly-shaped domain with proportional TC

- No-short selling and no-borrowing constraints:

$$(C_t, \delta_t) \in \mathcal{D}(W_t, \mathbf{x}_t) = \{(C, \delta) : \mathbf{x}_t + \delta \geq \mathbf{0} \text{ and } C/W_t + \mathbf{1}^\top (\mathbf{x}_t + \delta + \tau|\delta|) \leq 1\}$$

- No-aggregate leverage constraint:

$$(C_t, \delta_t) \in \mathcal{D}(\mathbf{x}_t) = \{(C, \delta) : \mathbf{1}^\top (\mathbf{x}_t + \delta) \leq 1\}$$

Recursive form and Bellman equation

1. For some economically motivated **terminal utility** function $U(\cdot, \cdot)$, let:

$$V_T(W_T, \mathbf{x}_T) := U(W_T, \mathbf{x}_T) \quad (3)$$

2. Define **recursively**, for any $t = T - 1, \dots, 0$:

$$V_t(W_t, \mathbf{x}_t) = \max_{(C_t, \boldsymbol{\delta}_t) \in \mathcal{D}(W_t, \mathbf{x}_t)} \{u(t, C_t) + \mathbb{E}[V(W_{t+1}, \mathbf{x}_{t+1}) | W_t, \mathbf{x}_t]\} , \quad (4)$$

where

$$(W_{t+1}, \mathbf{x}_{t+1}) = BC(W_t, \mathbf{x}_t, C_t, \boldsymbol{\delta}_t; R_f, \mathbf{R}_{t+1})$$

3. Obtain desired value function and optimal controls from last **iteration** at $t = 0$

Key issues for computational solution framework

1. "Exact" computation of $V_t(\cdot, \cdot)$ over entire domain $\mathcal{D}_t$ is infeasible/inappropriate
2. Need approximation of $V_t(\cdot, \cdot)$ from suitable set of "exact" computations $\{V_t(W_i, \mathbf{x}_i)\}_{i=1}^N$
 - Error in approximation of $V_t(\cdot, \cdot)$ needs to be quantifiable and "controllable"
 - Accuracy to bound error propagation in global solution as, e.g., D or T grows
 - Capture varying degrees of nonlinearity/nondifferentiability over domain $\mathcal{D}_t$
 - Efficiency with "exact" computations performed only within irregularly-shaped domain $\mathcal{D}_t$
3. Grid-free probabilistic approximation methods are preferable

Comprehensive Machine Learning Framework

- Based on recursive **grid-free probabilistic** approximation of value function $V_t(\cdot, \cdot)$
- Two **building blocks**:
 1. Accurate description of unknown **NTR**:

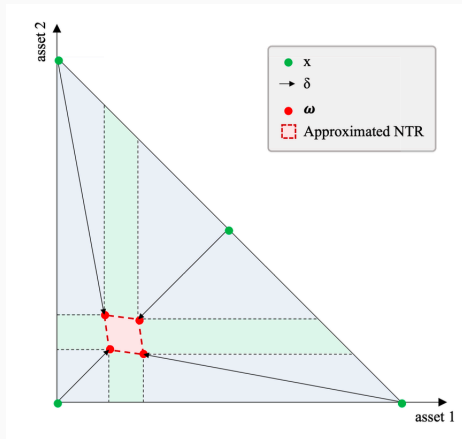
$$\Omega_t := \{(W_t, \mathbf{x}_t) \in \mathcal{D}_t : \delta_t^{opt} = 0\}$$

by means of suitable **approximate NTR** $\hat{\Omega}_t$

2. Accurate approximation of $V_t(\cdot, \cdot)|_{\hat{\Omega}_t}$ and $V_t(\cdot, \cdot)|_{\mathcal{D}_t \setminus \hat{\Omega}_t}$ using two different associated **Gaussian Process Regressions (GPRs)**

- Homothetic portfolio problem (e.g., $u(t, C) = \beta^t u(C)$ for a power utility with RRA $\gamma > 0$)
- Approximate NTR $\hat{\Omega}_t$ is a D -dimensional convex polytope with 2^D vertices
 - Covers most known NTR in the literature
 - NTR computations as convex hull of its vertices
- Computation of approximate NTR from 2^D computations of optimal portfolio policy

Approximate NTR II



- Sampling points $\{x_1, \dots, x_{2^D}\}$ in blue region identifies approximate NTR vertices as $\omega_i = x_i + \delta_i^{opt}$

Computation of t^{th} -period approximate NTR

Algorithm:

Input: $(t + 1)$ –period's (surrogate) value function $\mathcal{V}_{t+1}$

Data: Matrix $\tilde{\mathbf{X}}_t := \{\tilde{\mathbf{x}}_{ti}\}_{i=1}^N \in \mathcal{D}_t^N$ of $N = 2^D$ points

Output: Approximate NTR $\hat{\mathbf{\Omega}}_t$

1. **For** $i = 1$ **to** N **do:**

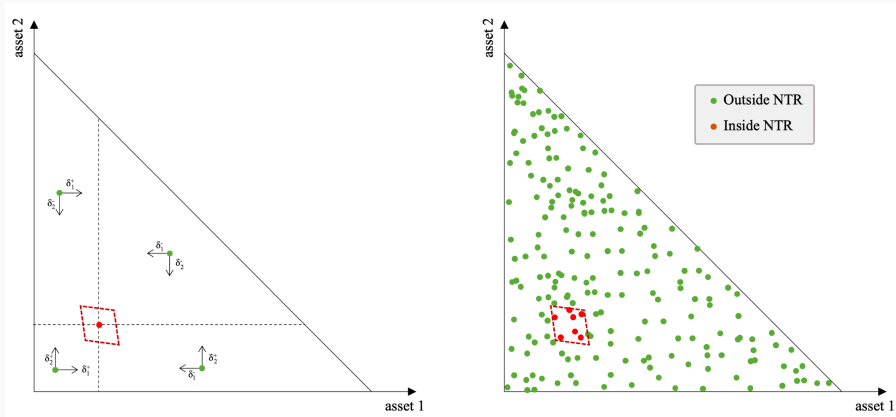
Compute vertex $\hat{\mathbf{\omega}}_{ti} := \tilde{\mathbf{x}}_{ti} + \hat{\delta}_{ti}^{\text{opt}}$ by solving Bellman equation (4) with $\mathcal{V}_{t+1}$ as next period's value function

end

2. Compute approximate NTR:

$$\hat{\mathbf{\Omega}}_t := \left\{ \sum_{i=1}^n \lambda_i \hat{\mathbf{\omega}}_{ti} : \sum_{i=1}^N \lambda_i = 1 \text{ and } \lambda_1, \dots, \lambda_N \geq 0 \right\}$$

Usefulness of approximate NTR for solving the portfolio problem



- Recovers **signs** of optimal portfolio policies
- Enables **heterogenous sampling** of value function training points inside/outside NTR
- Naturally identifies regions of value function **non-differentiability/high-nonlinearity**

Value function approximation with two GPRs I

- **Homothetic** portfolio choice problem (e.g., power utility): For any $t = T - 1, \dots, 0$,

$$V_t(W_t, \mathbf{x}_t) = W_t^{1-\gamma} v_t(\mathbf{x}_t) ; \gamma > 0$$

- **Bellman equation**: For any $t = T - 1, \dots, 0$:

$$v_t(\mathbf{x}_t) = \max_{(c_t, \boldsymbol{\delta}_t) \in \mathcal{D}(\mathbf{x}_t)} \left\{ u(c_t) + \beta \mathbb{E} \left[\pi_{t+1}^{1-\gamma} v_{t+1}(\mathbf{x}_{t+1}) | \mathbf{x}_t \right] \right\} \quad (5)$$

where $c_t := C_t/W_t$ and

$$\begin{aligned} \pi_{t+1} &:= (1 - c_t - \mathbf{1}^\top (\mathbf{x}_t + \boldsymbol{\delta}_t + \tau |\boldsymbol{\delta}_t|)) R_f + (\mathbf{x}_t + \boldsymbol{\delta}_t)^\top \mathbf{R}_{t+1} \\ \mathbf{x}_{t+1} &= \frac{1}{\pi_{t+1}} (\mathbf{x}_t + \boldsymbol{\delta}_t) \odot \mathbf{R}_{t+1} \end{aligned}$$

Value function approximation with two GPRs II

- Account for intrinsically **distinct** value function properties **inside/outside** NTR:

$$V_{1t} := v_t \mathbf{1}_{\hat{\Omega}_t} ; \quad V_{2t} := v_t \mathbf{1}_{\mathcal{D}_t \setminus \hat{\Omega}_t}$$

- "Disaggregated" Bellman equation: For any $t = T - 1, \dots, 0$:

$$v_{1t}(\mathbf{x}_t) + v_{2t}(\mathbf{x}_t) = \max_{(c_t, \delta_t) \in \mathcal{D}(\mathbf{x}_t)} \left\{ u(c_t) + \beta \mathbb{E} \left[\pi_{t+1}^{1-\gamma} (v_{1t+1}(\mathbf{x}_{t+1}) + v_{2t+1}(\mathbf{x}_{t+1})) \mid \mathbf{x}_t \right] \right\} \quad (6)$$

- Two distinct GPRs for approximating the value function **inside/outside** NTR:

$$\mathcal{GP}_{1t}(\mathbf{x}_t) + \mathcal{GP}_{2t}(\mathbf{x}_t) = \max_{(c_t, \delta_t) \in \mathcal{D}(\mathbf{x}_t)} \left\{ u(c_t) + \beta \mathbb{E} \left[\pi_{t+1}^{1-\gamma} (\mathcal{GP}_{1t+1}(\mathbf{x}_{t+1}) + \mathcal{GP}_{2t+1}(\mathbf{x}_{t+1})) \mid \mathbf{x}_t \right] \right\}$$

Gaussian process regression: Main Idea I

- **Supervised** machine learning method with **nonparametric Bayesian interpretation**
- Treats computed value function as **Gaussian process** with mean μ and kernel k :

$$v_{it}(\mathbf{x}) \sim \mathcal{GP}(\mu_{it}(\mathbf{x}), k_{it}(\mathbf{x}, \mathbf{x}'))$$

- For any $N \in \mathbb{N}$, Gaussian **finite sample (prior) distributions** of:

$$(v_{it}(\mathbf{x}_1), \dots, v_{it}(\mathbf{x}_N))$$

- For any $N, M \in \mathbb{N}$, closed-form Gaussian **conditional (posterior) distributions** of:

$$(v_{it}(\mathbf{x}'_1), \dots, v_{it}(\mathbf{x}'_M)) | (v_{it}(\mathbf{x}_1), \dots, v_{it}(\mathbf{x}_N))$$

Gaussian process regression: Main idea II

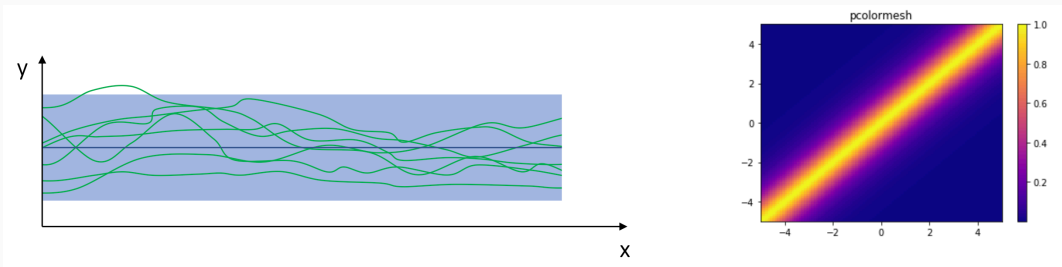
- Approximations of **unobserved** function values given set of **observed** function values:
 - **Point estimate** via $\mathbb{E}[(v_{it}(\mathbf{x}'_1), \dots, v_{it}(\mathbf{x}'_M)) | (v_{it}(\mathbf{x}_1), \dots, v_{it}(\mathbf{x}_N))]$
 - **Uncertainty measurement** via $\text{Cov}[(v_{it}(\mathbf{x}'_1), \dots, v_{it}(\mathbf{x}'_M)) | (v_{it}(\mathbf{x}_1), \dots, v_{it}(\mathbf{x}_N))]$
- $k(\mathbf{x}, \mathbf{x}')$ captures function values **similarity** in different regions of the domain:
 - **Matern kernel** for modelling pronounced function **nonlinearities**:

$$k_M(\mathbf{x}, \mathbf{x}') = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} d(\mathbf{x}, \mathbf{x}') \right)^\nu K_\nu \left(\sqrt{2\nu} d(\mathbf{x}, \mathbf{x}') \right)$$

with $d(\mathbf{x}, \mathbf{x}') := (\mathbf{x} - \mathbf{x}') [\text{diag}(\mathbf{l})]^{-2} (\mathbf{x} - \mathbf{x}')$

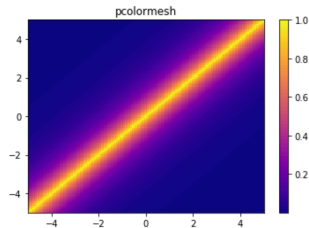
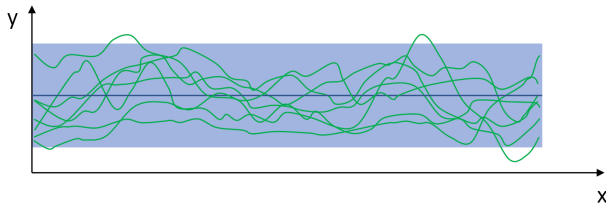
- **Hyper-parameters** obtained by maximising log likelihood over training sample

Gaussian process regression: Tunable smoothness with Matern kernel I



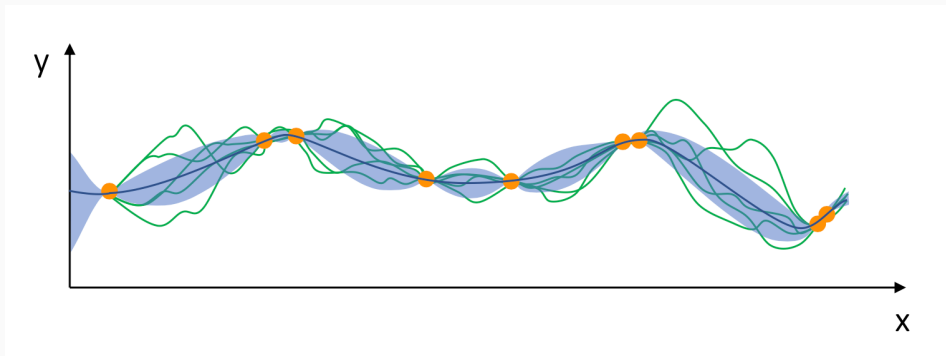
- Broader regions of **high similarity** model a priori **smoother** value function realizations
- **Training** of kernel hyperparameters allows **learning** of relevant degree of smoothness

Gaussian process regression: Tunable smoothness with Matern kernel II

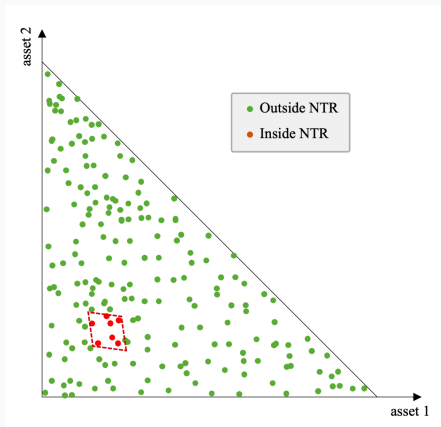


- Smaller regions of **high similarity** model a priori **less smooth** value function realizations
- **Training** of kernel hyperparameters allows **learning** of relevant degree of smoothness

Gaussian process regression: Posterior distributions



Putting things together



- Randomly generate **training samples** of points inside/outside NTR
- Train a **distinct GPR** on each subsample and **predict** remaining value function and policy values inside/outside NTR

Algorithm

Input: Terminal value function v_T , time horizon T , sampling size N

Output: Surrogate value functions $\{\mathcal{V}_{t-1}\}_{t=1}^T$ and approximate NTRs $\{\hat{\Omega}_{t-1}\}_{t=1}^T$

1. Set $\mathcal{V}_T = v_T$

2. **For** $t = T$ **to** 1 **do**:

Compute approximate NTR $\hat{\Omega}_{t-1}$ using $\mathcal{V}_t$ as next period's value function and sample N points $\mathbf{X}_{t-1} = \{\mathbf{x}_{t-1i}\}_{i=1}^N \in \mathcal{D}_{t-1}^N$

***For** $i = 1$ **to** N **do**:*

Obtain value function and policy values $(\hat{v}_{t-1i}, \hat{c}_{t-1i}, \hat{\delta}_{(t-1)i})$ for $\mathbf{x}_{t-1i}$ by solving Bellman equation (8) using $\mathcal{V}_t$ as next period's value function.

end

Fill training sets $\hat{\mathcal{D}}_{1t}, \hat{\mathcal{D}}_{2t}$ based on whether $\mathbf{x}_{t-1i} \in \hat{\Omega}_{t-1}$ or $\mathbf{x}_{t-1i} \notin \hat{\Omega}_{t-1}$ for some i

Learn from $\hat{\mathcal{D}}_{1t}, \hat{\mathcal{D}}_{2t}$ two GPR surrogates $\mathcal{V}_{1(t-1)}, \mathcal{V}_{2(t-1)}$ of $v_{1(t-1)}, v_{2(t-1)}$, to finally obtain a surrogate $\mathcal{V}_{t-1} = \mathcal{V}_{1(t-1)} + \mathcal{V}_{2(t-1)}$ of v_{t-1} .

end

Selected Findings

Findings I: Value function fit in benchmark IDD setting (2 assets)

N	Mean (%)	99.9 th percentile (%)	Max (%)
100	0.0070	0.101	0.110
300	0.0022	0.0372	0.0456
500	0.0007	0.0137	0.0159
700	0.0004	0.0061	0.0070

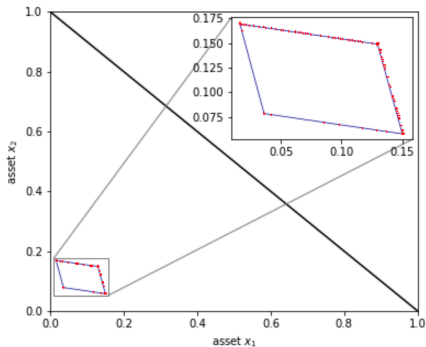
Table 3: Out-of-sample RAE value function fit summary statistics (in percent) of the final iteration (iteration 7) for various sample sizes. The summary statistics were computed using solutions from a model where the NTR was oversampled.

Findings II: Optimal policy fit in benchmark IDD setting (2-5 assets)

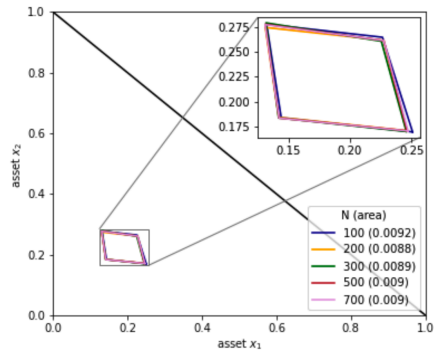
	2 risky assets		3 risky assets		5 risky assets	
N	500		700		2000	
REE (%)	mean	max	mean	max	mean	max
δ_1	0.0222	0.334	0.0341	0.2832	0.0377	0.2847
δ_2	0.0179	0.332	0.0308	0.2817	0.0306	0.3239
δ_3			0.0270	0.2833	0.0299	0.3831
δ_4					0.0294	0.2858
δ_5					0.0334	0.4485
c	1.0×10^{-8}	5.3×10^{-7}	1.0×10^{-8}	3.4×10^{-8}	4.4×10^{-6}	8.6×10^{-5}

Table 5: REE statistics for a high-dimensional portfolio optimization problem for final iteration.

Findings III: NTR fit in benchmark IDD setting



(a) NTR approximation error between computed policy and policy implied by approximated NTR.



(b) Comparing the final iteration's NTR computed using various sample sizes.

Directions of improvement and natural extensions

Natural dimensions of improvement/extension

- [Oversampling](#) of approximate NTR/NTR boundary
- [Bayesian reinforcement learning](#) exploiting GPR uncertainty quantification inside/outside NTR
- Gaussian-Hermite quadrature or Monte Carlo simulation for [integral approximations](#), depending on sampling size and state space dimension
- More plausible opportunity set and transaction cost [dynamics](#)

Extensions I: Stochastic opportunity set

- A stochastic opportunity set implies further **state variables** $\mathbf{z}_t$ in the **Bellman equation**
- $\mathbf{z}_t$ impacts the **conditional distribution** of returns given time t —information
- **Bellman equation**: For any $t = T - 1, \dots, 0$,

$$v_t(\mathbf{x}_t, \mathbf{z}_t) = \max_{(c_t, \boldsymbol{\delta}_t) \in \mathcal{D}(\mathbf{x}_t, \mathbf{z}_t)} \left\{ u(c_t) + \beta \mathbb{E} \left[\pi_{t+1}^{1-\gamma} v_{t+1}(\mathbf{x}_{t+1}, \mathbf{z}_{t+1}) | \mathbf{x}_t, \mathbf{z}_t \right] \right\} \quad (7)$$

where $c_t := C_t/W_t$ and

$$\begin{aligned} \pi_{t+1} &:= (1 - c_t - \mathbf{1}^\top (\mathbf{x}_t + \boldsymbol{\delta}_t + \tau |\boldsymbol{\delta}_t|)) R_f + (\mathbf{x}_t + \boldsymbol{\delta}_t)^\top \mathbf{R}_{t+1} \\ \mathbf{x}_{t+1} &= \frac{1}{\pi_{t+1}} (\mathbf{x}_t + \boldsymbol{\delta}_t) \odot \mathbf{R}_{t+1} \end{aligned}$$

- $\mathbf{z}_t$ —state dynamics may be defined over an **irregularly-shaped** state space in, e.g., **regime-switching/mixture models**

Extensions II: Stochastic asset specific transaction costs

- Stochastic frictions may imply further **state variables** ($\mathbf{z}_t, W_t$) in the **Bellman equation**
- They impact the **budget constraint** of the portfolio problem
- **Bellman equation**: For any $t = T - 1, \dots, 0$,

$$v_t(\mathbf{x}_t, \mathbf{z}_t, W_t) = \max_{(c_t, \boldsymbol{\delta}_t) \in \mathcal{D}(\mathbf{x}_t)} \left\{ u(c_t) + \beta \mathbb{E} \left[\pi_{t+1}^{1-\gamma} v_{t+1}(\mathbf{x}_{t+1}, \mathbf{z}_{t+1}, W_{t+1}) | \mathbf{x}_t, \mathbf{z}_t, W_t \right] \right\}$$

where $c_t := C_t/W_t$ and

$$\pi_{t+1} := (1 - c_t - \mathbf{1}^\top (\mathbf{x}_t + \boldsymbol{\delta}_t + \boldsymbol{\tau}'(\mathbf{x}_t, \mathbf{z}_t, W_t) |\boldsymbol{\delta}_t|)) R_f + (\mathbf{x}_t + \boldsymbol{\delta}_t)^\top \mathbf{R}_{t+1}$$

$$\mathbf{x}_{t+1} = \frac{1}{\pi_{t+1}} (\mathbf{x}_t + \boldsymbol{\delta}_t) \odot \mathbf{R}_{t+1}$$

- Transaction cost function may even be defined on **irregularly-shaped** domain and feature **nonlinearities/nondifferentiability** (e.g., with **threshold-type** structures)

Two Regimes / Two Assets Model with Stochastic Transaction Costs

Table 1: Regime-dependent parameters follow a two-regime Markov Chain; stochastic transaction costs average 1% per asset across regimes.

	Low volatility ($z = 0$)	High volatility ($z = 1$)
Mean log returns $\mu(z)$	(0.0741, 0.0648)	(0.0817, 0.0772)
Volatilities of log returns $\sigma(z)$	(0.1361, 0.0860)	(0.2171, 0.1949)
Correlation of log returns $\rho(z)$	0.293	0.316
Transaction costs $\tau(z)$ (%)	(0.40, 0.70)	(1.60, 1.30)
Transition probabilities $P[z, \cdot]$	(0.656, 0.344)	(0.345, 0.655)

Properties of No Trade Regions

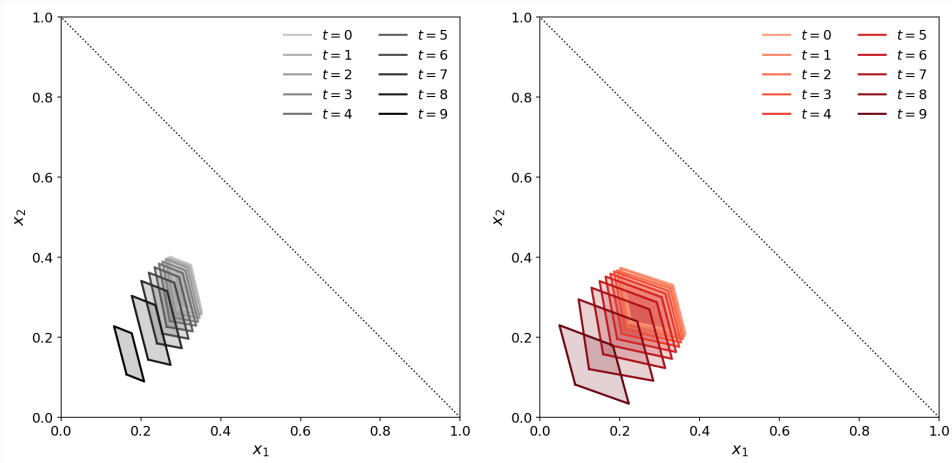


Figure 1: No trade region. Horizon patterns of the NTR in the low (left) and the high (right) volatility regime.

Realized portfolio patterns

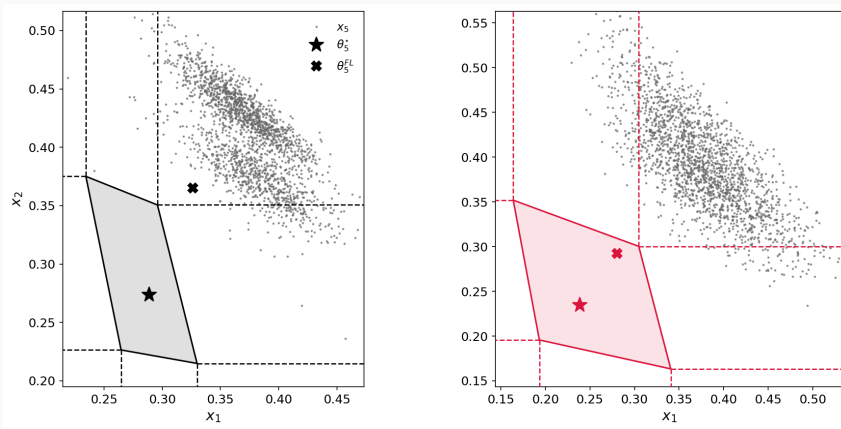


Figure 2: Realized optimal portfolios. Simulated states at $t = 5$ (grey points) in low-volatility (left) and high-volatility (right) regimes. NTRs define lines identifying different policy types, while stars (crosses) highlight target (frictionless) optimal portfolios $\theta_5^*(z)$ ($\theta_5^{FL}(z)$).

Myopic vs. intertemporal hedging portfolios

	x		$x + \delta_0^*(x)$		$x + \delta_{T-1}^*(x)$		HD	
state	x_1	x_2	x_1	x_2	x_1	x_2	x_1	x_2
<i>Low volatility ($z = 0$)</i>								
all bond	0.000	0.000	0.297	0.274	0.164	0.107	0.133	0.167
all asset 1	1.000	0.000	0.353	0.263	0.208	0.090	0.145	0.173
all asset 2	0.000	1.000	0.273	0.401	0.133	0.228	0.140	0.173
fully invested	0.500	0.500	0.326	0.381	0.177	0.210	0.149	0.171
$\theta_0^*(0)$	0.312	0.330	0.312	0.330	0.178	0.211	0.134	0.119
<i>High volatility ($z = 1$)</i>								
all bond	0.000	0.000	0.228	0.238	0.088	0.082	0.140	0.156
all asset 1	1.000	0.000	0.365	0.208	0.223	0.034	0.142	0.174
all asset 2	0.000	1.000	0.203	0.373	0.048	0.230	0.155	0.143
fully invested	0.500	0.500	0.333	0.331	0.182	0.180	0.151	0.151
$\theta_0^*(1)$	0.282	0.288	0.282	0.288	0.183	0.181	0.099	0.107

Cost of illiquidity

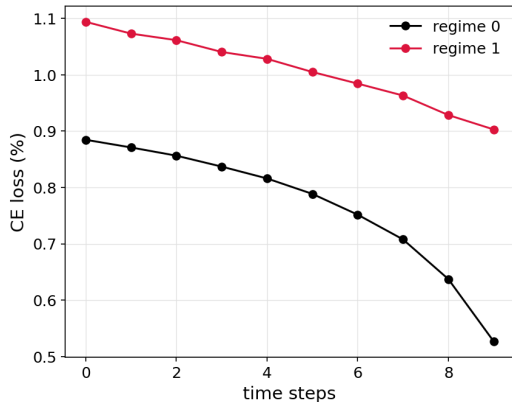
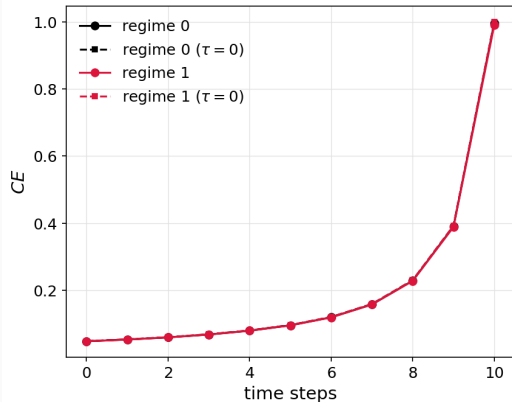


Figure 3: Certainty-equivalent wealth. Left: average $CE_t(x, z)$ across regimes in markets with and without transaction costs. Right: relative CE loss across regimes between markets with and without transaction costs.

Control of error propagation

Table 2: Policy OOS Relative Euler Error and GP fit Relative Absolute Error (1000 points).

		mean	median	q_{90}	max
<i>Panel A: first iteration ($t = 9$)</i>					
OOS RAE		5.0×10^{-9} (7.9×10^{-10})			
REE on δ^+	$z = 0$	2.8×10^{-5}	4.3×10^{-6}	3.6×10^{-5}	1.7×10^{-3}
	$z = 1$	3.3×10^{-4}	5.5×10^{-5}	9.3×10^{-4}	3.7×10^{-3}
REE on c	$z = 0$	8.1×10^{-12}	5.4×10^{-13}	2.5×10^{-11}	1.0×10^{-10}
	$z = 1$	6.4×10^{-12}	$< 10^{-15}$	2.0×10^{-11}	1.0×10^{-10}
<i>Panel B: last iteration ($t = 0$)</i>					
OOS RAE		1.7×10^{-5} (3.2×10^{-6})			
REE on δ^+	$z = 0$	1.5×10^{-6}	8.7×10^{-7}	4.1×10^{-6}	1.5×10^{-5}
	$z = 1$	2.9×10^{-6}	1.3×10^{-6}	7.4×10^{-6}	2.7×10^{-5}
REE on c	$z = 0$	1.4×10^{-9}	5.6×10^{-12}	1.7×10^{-10}	1.5×10^{-7}
	$z = 1$	1.6×10^{-9}	2.0×10^{-12}	1.6×10^{-10}	1.9×10^{-7}

Comparison with grid-search dynamic programming I

Table 3: GP vs. grid-search dynamic programming with linear interpolation (Lynch and Tan (2010, 2011)), at an equal point budget (grid spacing $h = 1/44$) and at their resolution ($h = 0.01$).

	GP	grid ($h = 1/44$)	grid ($h = 0.01$)
Solved points per (t, z)	1,000	1,035	5,151
OOS RAE	$5 \times 10^{-9} / 4.2 \times 10^{-5}$	$1.6 \times 10^{-5} / 2.0 \times 10^{-5}$	$7.3 \times 10^{-6} / 9.5 \times 10^{-6}$
Rel. Euler error on δ	1.9×10^{-5}	1.5×10^{-4}	6.6×10^{-5}
Rel. Euler error on c	4.6×10^{-7}	3.9×10^{-5}	4.0×10^{-5}
Allocation error, $\tau > 0$ (max)	0.0004	0.006	0.003
No-trade area, ratio to GP	—	0.84–1.09	1.02–1.09
NTR avg. disagreement (%)	—	5.4	5.1
CE-loss overstatement (bps)	—	1.5–4.9	1.1–4.2

Comparison with grid-search dynamic programming II

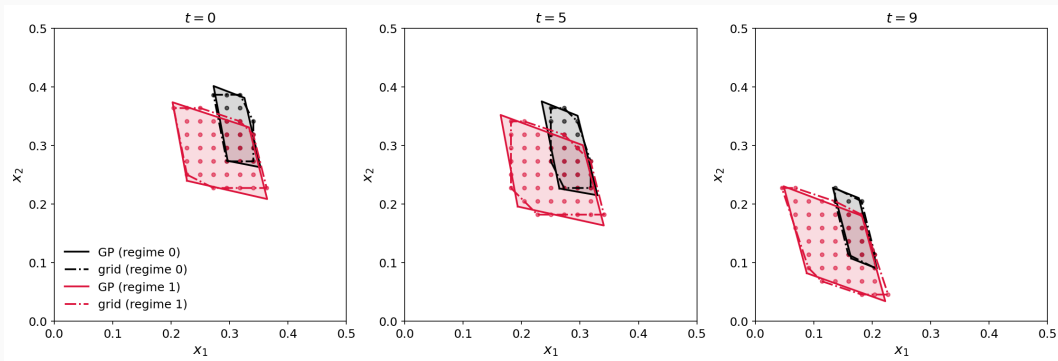


Figure 4: NTRs at $t \in \{0, 5, 9\}$: GP-based $\hat{\Omega}_t(z)$ (solid, shaded) against the NTR of the equal-budget grid method (dots) and its convex hull (dash-dot).

Extensions to very high-dimensional asset space

- Gaussian process approximations with [active subspaces](#) (Bilionis Scheidegger (2019)):
 - Reduce [model dimension](#) via a projection on lower-dimensional linear manifold of the input space having [maximal response](#) variation
- [Geometric deep learning](#) (Bronstein Bruna Cohen Veličković (2021)):
 - Exploit [geometric model properties](#) such as symmetries, to build [improved/scalable](#) learning architectures
- [Scalable deep kernels](#) (Wilson Hu Salakhutdinov Xing (2016)):
 - Combine structural properties of deep learning with nonparametric flexibility of kernels
 - Closed-form kernels as [drop-in replacements](#) for standard kernels, with benefits in [expressive power](#) and [scalability](#)

Conclusions

Conclusions

Scalable machine learning solution framework for dynamic portfolio choice with TC:

- Accomodating nonlinearities, nondifferentiabilities and irregularly-shaped domains
- "Tunable-smooth" and allows uncertainty quantification

Relies on accurate probabilistic approximations of value and policy functions:

- A novel accurate approximation of the irregularly-shaped NTR
- Two accurate GPR approximations of the value function inside/outside the NTR
- Bayesian reinforcement learning

Thank you!